KETI 한국전자기술연구원
Korea Electronics Technology Institute

성균관대학교
SUNG KYUN KWAN UNIVERSITY

Interface to In-Network Computing Functions for Cooperative Intelligent Transportation Systems

(draft-an-nmrg-i2icf-cits-01)

March 16, 2026
Shenzhen in China

Byoungman Robert An, and Jaehoon Paul Jeong

Email : bman@keti.re.kr, pauljeong@skku.edu

IETF

I2ICF Side Meeting

Motivation of This Draft

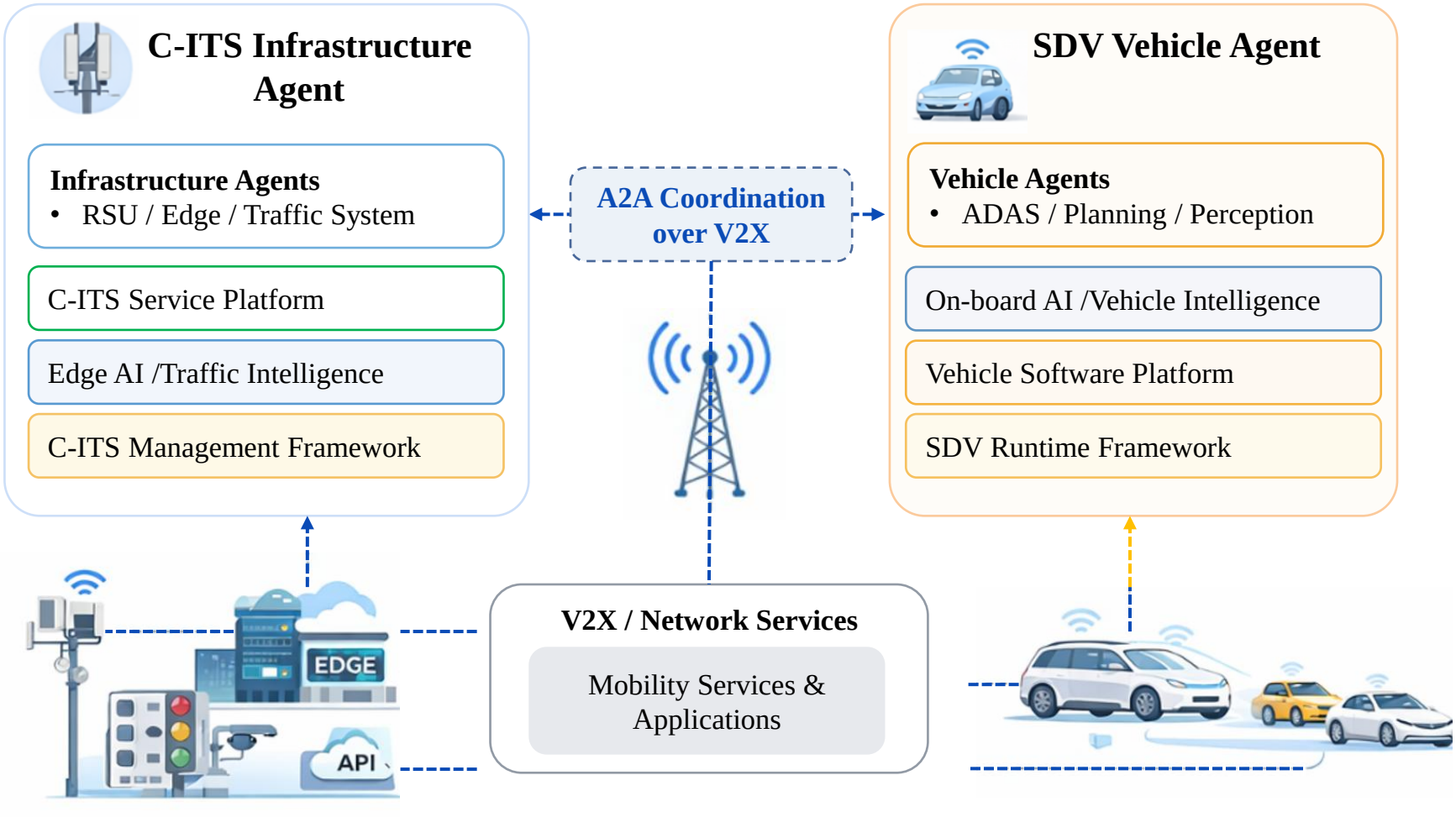
❖ draft-an-nmrg-i2icf-cits-01

- Link : https://datatracker.ietf.org/doc/draft-an-nmrg-i2icf-cits/
- Explores I2ICF for Agent-to-Agent (A2A) collaboration in C-ITS and SDV environments.
- Defines a framework for orchestrating, managing, and monitoring in-network functions for C-ITS.
- Vehicles, infrastructure, and network nodes operate as cooperative software agents in future C-ITS.
- A2A communication enables distributed decision-making among vehicle, edge, and network agents.
- I2ICF supports real-time optimization of V2X communication and traffic services at the network edge.

❖ Main contents of this draft

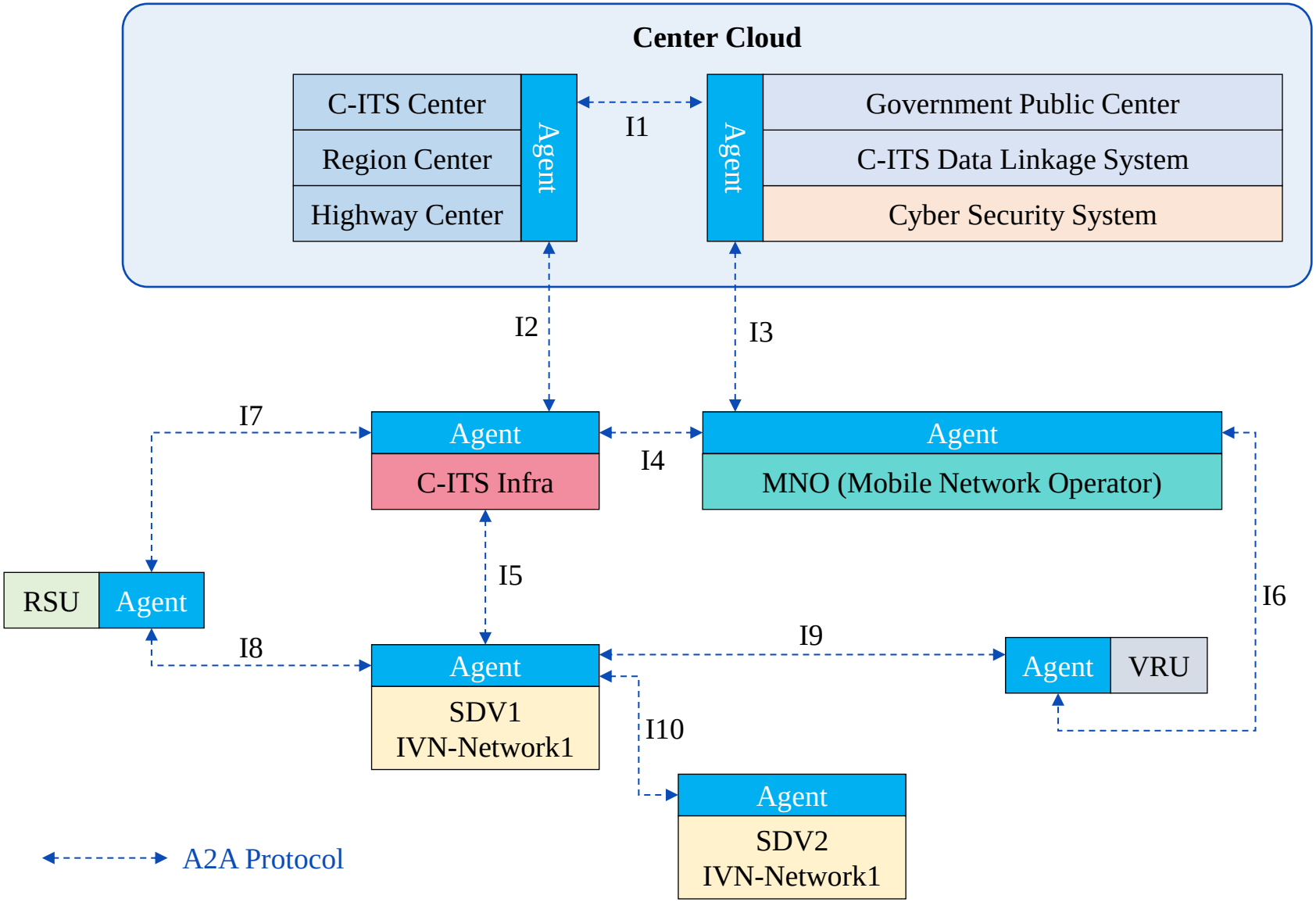
- Intent-Based Management for multi-agent C-ITS systems
- A2A collaboration over programmable networks
- Architecture and interfaces for I2ICF orchestration
- Use cases in SDV and cooperative mobility services
- Security and trust considerations for agent-based C-ITS

A2A Collaboration Architecture for C-ITS and SDV



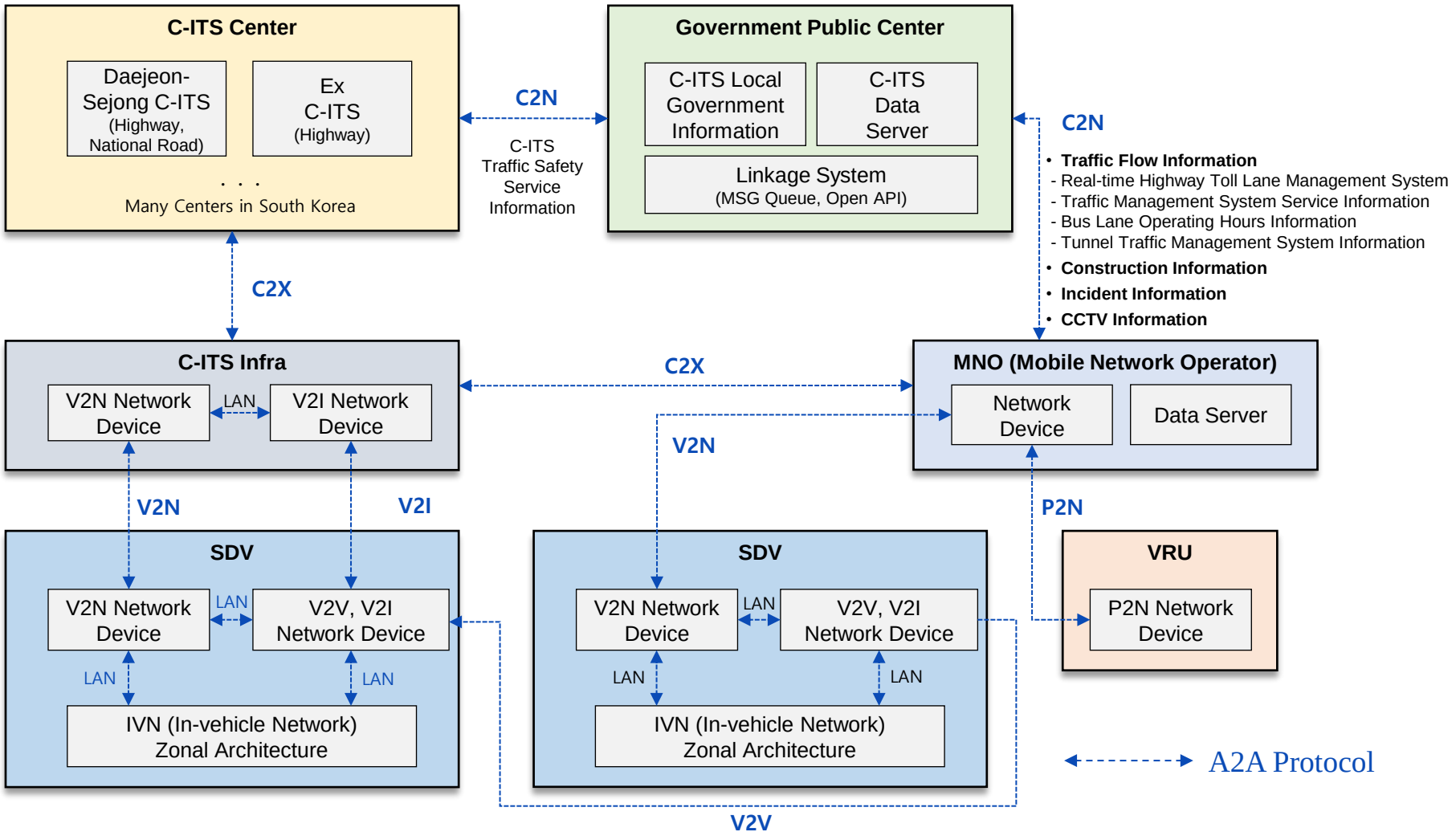
< Source : Google >

A2A Framework and Interfaces for C-ITS and SDV



A2A Use Cases of I2ICF for C-ITS

A2A Data Processing, Streamline Traffic Management



Summary and Next Steps

❖ Summary

- Proposes a structured framework for I2ICF registration, configuration, management, and monitoring.
- Evaluates C-ITS use cases, enabling A2A orchestration for V2X networks, seamless integration with C-ITS, and interoperability with MNOs.
- Highlights the role of I2ICF in strengthening cybersecurity across interconnected vehicular ecosystems.
- Introduces an Intent-Based Management architecture for C-ITS with agentic AI and A2A coordination.

❖ Next Steps

- This draft will be extended with I2ICF-based C-ITS and SDV use cases, including:
  - A2A-based C-ITS networking with agentic AI
  - Real-time SDV data processing and coordination
  - End-to-end cooperative driving via A2A communication
  - Enhanced cybersecurity for C-ITS and MNO integration

Unframed Perspective

"틀에서 벗어난 시각으로 미래를 이끌어 간다"

KETI Korea Electronics Technology Institute

This work was supported by Institute of Information & Communications Technology Planning & Evaluation (IITP) grant funded by the Korea Ministry of Science and ICT (MSIT) (No. 2022-0-00199, 5G-NR-V2X performance verification for connected Autonomous Driving).
This work was in part supported by Institute of Information & Communications Technology Planning & Evaluation (IITP) grant funded by the Korea Ministry of Science and ICT (MSIT) (No. RS-2024-00398199 and RS-2022-II221015).